1 Physics-aware encoding operator and fast gradients

We reconstruct images x on a grid $\Omega \subset \mathbb{R}^3$ from whitened multi-coil spiral data, without density compensation. The operator A contains off-resonance, R_2^*, fat, the concomitant field of the whole gradient history, gradient nonlinearity (GNL) and cardiac motion between readouts. It costs $L \approx 10$ FFT pairs per coil. Everything that is not a smooth function of the readout time is absorbed into a per-sample *chirp-aware stencil* (Section 1.3). Section 1.5 puts the velocity parametrisation on top of A, Section 1.6 gives the cost of one gradient $A^{\mathsf{H}}(Ax - y)$ and the levers that reduce it, and Section 1.7 gives the algebra of $\mathsf{N} = A^{\mathsf{H}}A$.

1.1 Signal model and phase split

With $\boldsymbol{r}$ the true scanner position of a voxel (after GNL, including the FOV offset), S_c the whitened coil maps, $\boldsymbol{k}_j$ the nominal trajectory and τ_j the time since excitation,

$$y_{cj} = \sum_{\boldsymbol{r}\in\Omega} S_c(\boldsymbol{r})\, x(\boldsymbol{r})\, \mathrm{e}^{-\mathrm{i}\Theta_j(\boldsymbol{r})} + n_{cj}, \qquad \Theta_j(\boldsymbol{r}) = \boldsymbol{k}_j\cdot\boldsymbol{r} + \big(\omega(\boldsymbol{r}) - \mathrm{i}R_2^*(\boldsymbol{r})\big)\tau_j + \sum_{q=1}^{4} w_{jq}\, m_q(\boldsymbol{r}), \quad (1)$$

with $n \sim \mathcal{CN}(0, I)$ after whitening by $\Psi^{-1/2}$. The concomitant monomials are $m = (z^2,\ x^2{+}y^2,\ xz,\ yz)$. A field with $\nabla\cdot\boldsymbol{B} = \nabla\times\boldsymbol{B} = 0$ has $\boldsymbol{B}_\perp = (G_x z - \frac{1}{2}G_z x,\ G_y z - \frac{1}{2}G_z y)$ and $|\boldsymbol{B}| \approx B_z + |\boldsymbol{B}_\perp|^2/2B_0$, so [2, 1]

$$\sum_q w_{jq} m_q(\boldsymbol{r}) = \frac{\gamma}{2B_0}\big\langle W_j,\ M_{\boldsymbol{r}}^\top M_{\boldsymbol{r}}\big\rangle, \qquad W_j = \int_0^{t_j} \boldsymbol{G}\boldsymbol{G}^\top \mathrm{d}t, \qquad M_{\boldsymbol{r}} = \begin{pmatrix} z & 0 & -x/2 \\ 0 & z & -y/2 \end{pmatrix}, \quad (2)$$

with relative error $O(|\boldsymbol{B}_\perp|^2/B_0^2) \approx 2\cdot 10^{-4}$ at 3 T. Here $\boldsymbol{G}$ is the physical gradient (after rotation and FOV offset) and W_j integrates from excitation, so the bipolars, the k_z blip and the readout all contribute.

Split. Let $\bar{w}_q(\tau)$ be the mean of w_{jq} over the readouts of an encode at ADC time τ, and $\Delta w_{jq} = w_{jq} - \bar{w}_q(\tau_j)$. Then

$$\Theta_j(\boldsymbol{r}) = \boldsymbol{k}_j\cdot\boldsymbol{r} + \psi(\boldsymbol{r}, \tau_j) + \Delta_j(\boldsymbol{r}), \qquad \psi(\boldsymbol{r}, \tau) = \big(\omega - \mathrm{i}R_2^*\big)\tau + \sum_q \bar{w}_q(\tau) m_q, \qquad \Delta_j = \sum_q \Delta w_{jq} m_q. \quad (3)$$

ψ is an arbitrary spatial function of $(\boldsymbol{r}, \tau)$ and is carried exactly by time segments (Section 1.2). Δ_j is small and polynomial. In logical coordinates $\boldsymbol{\rho}$ (GNL-warped positions $\boldsymbol{r}(\boldsymbol{\rho})$) we fit each $m_q(\boldsymbol{r}(\boldsymbol{\rho}))$ over the support with monomials of degree ≤ 4:

$$\Delta_j(\boldsymbol{\rho}) = \phi_{0j} + \boldsymbol{b}_j\cdot\boldsymbol{\rho} + \boldsymbol{\rho}^\top \Delta C_j\, \boldsymbol{\rho} + c_j(\boldsymbol{\rho}) + \varepsilon_j(\boldsymbol{\rho}), \qquad c_j(\boldsymbol{\rho}) = \sum_q \Delta w_{jq} \sum_{3\le|\alpha|\le 4} h_{q\alpha}\boldsymbol{\rho}^\alpha. \quad (4)$$

The constant ϕ_{0j} becomes a sample phase and the linear part an exact shift $\boldsymbol{k}'_j = \boldsymbol{k}_j + \boldsymbol{b}_j$. The chirp $\chi_j = \boldsymbol{\rho}^\top \Delta C_j \boldsymbol{\rho} + c_j$ goes into the stencil. Phase budget of the full concomitant field, simulated 3 T spiral (80 mT/m, 36 cm FOV, 4.4 ms readout, 3 cm off-centre) [measured]:

	constant ϕ_0	linear $\boldsymbol{b}\cdot\boldsymbol{\rho}$	quadratic (corners)
max over samples and FOV	0.08 rad	0.96 rad (≤ 0.29 cells)	3.1 rad

Without GNL the quadratic fit residual ε_j is zero. At GNL-warped positions it is $4.5\cdot 10^{-3}$ rad rms (0.056 max) for degree 2, and degree 4 removes most of it (table in Section 1.3).

1.2 Time segments

With segment diagonals $D_\ell(\boldsymbol{r}) = \mathrm{e}^{-\mathrm{i}\psi(\boldsymbol{r},\tau_\ell)}$, $\ell = 1..L$, the interpolators are the least-squares solution over the support Ω_s [3]:

$$\begin{aligned} \boldsymbol{a}(\tau) = \operatorname*{arg\,min}_{\boldsymbol{a}\in\mathbb{C}^L} \sum_{\boldsymbol{r}\in\Omega_s} \left| \mathrm{e}^{-\mathrm{i}\psi(\boldsymbol{r},\tau)} - \sum_\ell a_\ell D_\ell(\boldsymbol{r}) \right|^2 \iff G\boldsymbol{a}(\tau) = \boldsymbol{\beta}(\tau), \\ G_{\ell\ell'} = \langle D_\ell, D_{\ell'} \rangle_{\Omega_s}, \qquad \beta_\ell(\tau) = \langle D_\ell, \mathrm{e}^{-\mathrm{i}\psi(\cdot,\tau)} \rangle_{\Omega_s}. \end{aligned} \tag{5}$$

$\boldsymbol{a}$ is tabulated at every ADC sample, which removes a 3–7 $\cdot 10^{-4}$ table floor. The rank is set by the number of phase cycles $\Delta f\, T_{\mathrm{ro}}$ (Δf the 1–99 % off-resonance span), not by resolution. Default: $L = \lceil 1.3\, \Delta f\, T_{\mathrm{ro}} + 3.5 \rceil$. Model error against an f64 direct sum, smooth 450 Hz field [measured]:

L	2 ms	4 ms	8 ms	12 ms
4	$9 \cdot 10^{-4}$			
6	$8 \cdot 10^{-6}$	$2.5 \cdot 10^{-4}$	$1.5 \cdot 10^{-2}$	
8			$6 \cdot 10^{-4}$	
10			$2.7 \cdot 10^{-5}$	$8.9 \cdot 10^{-4}$

Optimal basis. For fixed diagonals the interpolators Equation (5) are already least-squares optimal at every τ, so a better model can only come from the diagonals. Minimising $\sum_{\boldsymbol{r}\in\Omega_s} \sum_i |g(\boldsymbol{r},\tau_i) - \sum_\ell D_\ell(\boldsymbol{r}) a_\ell(\tau_i)|^2$ over both is a truncated SVD of $G = [g(\boldsymbol{r},\tau_i)]$ (Eckart–Young): with V_L the leading right singular vectors,

$$D_\ell(\boldsymbol{r}) = \sum_i g(\boldsymbol{r},\tau_i)\, V_{i\ell}, \qquad \boldsymbol{a}(\tau) \text{ from Equation (5) with these } D_\ell, \tag{6}$$

optionally weighted by the expected signal energy per voxel. The diagonals are complex and the operator cost is unchanged. They are built once per encode, not per frame: ψ depends on the readout time and the field, not on the frame's trajectory. Separable-model error on the simulator's field map (2.12 phase cycles, 3.56 ms; rms / worst τ) [measured, CPU]:

L	6	7	8	10
fixed times	$5.6 \cdot 10^{-3}$ / $1.1 \cdot 10^{-2}$	$2.0 \cdot 10^{-3}$ / $4.5 \cdot 10^{-3}$	$6.0 \cdot 10^{-4}$ / $1.5 \cdot 10^{-3}$	$3.5 \cdot 10^{-5}$ / $1.1 \cdot 10^{-4}$
SVD basis	$3.6 \cdot 10^{-3}$ / $9.4 \cdot 10^{-3}$	$1.3 \cdot 10^{-3}$ / $3.3 \cdot 10^{-3}$	$3.5 \cdot 10^{-4}$ / $9.5 \cdot 10^{-4}$	$1.4 \cdot 10^{-5}$ / $3.4 \cdot 10^{-5}$

At equal L the SVD basis has about 0.6× the error, worth roughly half a segment. The rank itself is bounded by the time–bandwidth product $\Delta f\, T_{\mathrm{ro}}$, so no separable $\sum_\ell D_\ell(\boldsymbol{r}) a_\ell(\tau)$ needs fewer terms than the SVD. The segments are not tied to the concomitant handling: the stencils only see the per-sample deviation from ψ, whatever basis represents ψ.

Species and decay in the diagonals. Complex ψ carries R_2^* at no cost. Water and fat with a known fat fraction $f(\boldsymbol{r})$ and the six-peak spectrum $c(\tau)$ share one set of segments:

$$D_\ell(\boldsymbol{r}) = g_{\boldsymbol{r}}(\tau_\ell), \qquad g_{\boldsymbol{r}}(\tau) = \mathrm{e}^{-\mathrm{i}\psi(\boldsymbol{r},\tau)} \big(1 - f(\boldsymbol{r}) + f(\boldsymbol{r})\, c(\tau)\big). \tag{7}$$

With this, $L = 12$ gives $5.0 \cdot 10^{-4}$ and $L = 14$ gives $6.3 \cdot 10^{-5}$. Two species with separate segments ($2L = 16$ paths) give $6.5 \cdot 10^{-4}$ [measured, CPU].

1.3 Chirp-aware stencils

After Equations (3) and (4), with $v_{c\ell} = S_c D_\ell x$, sample j needs the *modulated row*

$$(Ax)_{cj} = \mathrm{e}^{-\mathrm{i}\phi_{0j}} \sum_\ell a_\ell(\tau_j) \sum_{\boldsymbol{r}} v_{c\ell}(\boldsymbol{r})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{k}'_j\cdot\boldsymbol{r}}\, \mathrm{e}^{-\mathrm{i}\chi_j(\boldsymbol{r})}. \tag{8}$$

The NUFFT represents a plane wave on Ω through the oversampled grid $\boldsymbol{u} \in \mathbb{Z}^3/(\sigma N)$ and a window φ with apodisation $\hat{\varphi}$ [5, 6]: $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} = \hat{\varphi}(\boldsymbol{r})^{-1} \sum_{\boldsymbol{u}} \varphi(\boldsymbol{k}-\boldsymbol{u})\mathrm{e}^{-\mathrm{i}\boldsymbol{u}\cdot\boldsymbol{r}} + \epsilon$. A modulated row needs a different set of taps:

Definition 1 (chirp-aware stencil)**.** $\Gamma_j : \mathbb{Z}^3 \to \mathbb{C}$*, supported on* $\|\boldsymbol{u} - \boldsymbol{k}'_j\|_\infty \le J/2$*, with*

$$\sum_{\boldsymbol{u}} \Gamma_j(\boldsymbol{u})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{u}\cdot\boldsymbol{r}} \;\approx\; \hat{\varphi}(\boldsymbol{r})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{k}'_j\cdot\boldsymbol{r}}\, \mathrm{e}^{-\mathrm{i}\chi_j(\boldsymbol{r})}, \qquad \boldsymbol{r} \in \Omega. \tag{9}$$

Then, with $G_{c\ell} = \mathcal{F}_\sigma[\hat{\varphi}^{-1} S_c D_\ell x]$ (one oversampled FFT per channel),

$$(Ax)_{cj} = \mathrm{e}^{-\mathrm{i}\phi_{0j}} \sum_{\boldsymbol{u}} \Gamma_j(\boldsymbol{u}) \sum_\ell a_\ell(\tau_j)\, G_{c\ell}(\boldsymbol{u}), \qquad \big(A^{\mathsf{H}} y\big) = \sum_c \overline{S_c} \sum_\ell \overline{D_\ell}\, \hat{\varphi}^{-1} \mathcal{F}_\sigma^{\mathsf{H}} \Big[\sum_j \overline{\Gamma_j}\, \overline{a_\ell(\tau_j)}\, \mathrm{e}^{\mathrm{i}\phi_{0j}} y_{cj} \Big]. \tag{10}$$

Γ_j *depends on neither the coil nor the segment.* It is evaluated once per (sample, tap) and reused for all $C \cdot L$ channels. A^{H} is the exact transpose. A plain NUFFT has $\Gamma_j = \varphi(\boldsymbol{k}'_j - \cdot)$. We use two constructions of Γ_j.

(a) Taylor stencils: derivative weights of the window (production)

From $\rho^\alpha \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\rho}} = (\mathrm{i}\partial_{\boldsymbol{k}})^\alpha \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\rho}}$, any polynomial phase acts on the plane wave as a differential operator in $\boldsymbol{k}$:

$$\mathrm{e}^{-\mathrm{i}\chi(\boldsymbol{\rho})}\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\rho}} = \sum_{n=0}^{P} \frac{(-\mathrm{i})^n}{n!}\, \chi(\mathrm{i}\partial_{\boldsymbol{k}})^n\, \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\rho}} + R_P, \qquad |R_P| \le \frac{|\chi|^{P+1}}{(P+1)!}. \tag{11}$$

Applied to the interpolant this gives the stencil in closed form:

Proposition 2 (Taylor stencil)**.** *Let* $\Delta C : \nabla\nabla = \sum_{pq} \Delta C_{pq} \partial_p \partial_q$*. Then*

$$\Gamma_j^{(P)}(\boldsymbol{u}) = \Big[1 + \mathrm{i}\, \Delta C_j : \nabla\nabla - \tfrac{1}{2}\big(\Delta C_j : \nabla\nabla\big)^2 - \mathrm{i} \sum_{3 \le |\alpha| \le 4} \mathrm{i}^{|\alpha|} \Big(\sum_q \Delta w_{jq} h_{q\alpha} \Big) \partial^\alpha \Big] \varphi(\boldsymbol{\kappa} - \boldsymbol{u}) \Big|_{\boldsymbol{\kappa} = \boldsymbol{k}'_j} \tag{12}$$

($P = 2$ uses the first three terms, and order 3 adds the last term to first order). The result is the exact derivative of the interpolant $\sum_{\boldsymbol{u}} \varphi(\boldsymbol{k}-\boldsymbol{u}) G(\boldsymbol{u})$*, not a finite difference. The error is* $|\chi_j|^{P+1}/(P+1)!$ *plus the window error of* $\partial^\alpha \varphi$*.*

Proof. $\chi = \boldsymbol{\rho}^\top \Delta C \boldsymbol{\rho} \Rightarrow \chi(\mathrm{i}\partial) = -\Delta C : \nabla\nabla$. Insert into Equation (11) for $n \le 2$, use $\rho^\alpha \mapsto \mathrm{i}^{|\alpha|}\partial^\alpha$ for c_j, and move $\partial_{\boldsymbol{k}}$ onto $\varphi(\boldsymbol{k}-\boldsymbol{u})$. (Checked against the exact chirped sum.) □

Separable evaluation. $\partial^\alpha \varphi(\boldsymbol{\kappa} - \boldsymbol{u}) = \prod_a \varphi^{(\alpha_a)}(\kappa_a - u_a)$, so

$$\Gamma_j^{(P)}(\boldsymbol{u}) = \sum_{|\alpha| \le 4} \gamma_{j\alpha} \prod_{a=1}^{3} \varphi^{(\alpha_a)}(\kappa_{ja} - u_a), \qquad \gamma_j \in \mathbb{C}^{35}. \tag{13}$$

Each tap is a sum of 35 rank-one products of 1-D tables $\varphi^{(0..4)}$ (5 values per axis per tap). The coefficients come from the 6 numbers of ΔC_j, their 15 distinct fourth-order products, and $u_j \in \mathbb{R}^4$ with a 4×25 table h. Everything is formed in registers. *Window.* Order 2 needs $\varphi^{(4)}$ bounded. The exponential of semicircle $e^{\beta(\sqrt{1-z^2}-1)}$ has $\sqrt{\cdot}$-singular derivatives at $|z| = 1$, so they are clamped to 0 for $|z| > 0.98$, where $\varphi < 10^{-7}$; without the clamp the fourth-derivative error is $46\times$ larger. Interpolated value and ∂^2 errors (1-D) [measured]:

σ (grid volume vs. σ=2)	W	value	∂^2
2.0 (1.00)	7	$9 \cdot 10^{-7}$	$8 \cdot 10^{-5}$
1.5 (0.42)	10	$6 \cdot 10^{-8}$	$6 \cdot 10^{-5}$
1.25 (0.24)	10	$2 \cdot 10^{-6}$	$3 \cdot 10^{-5}$

The correction term is $\lesssim 0.2$ rad, so ∂^2 accuracy 10^{-3} suffices: $\sigma = 1.25$, $W = 10$. *Accuracy* vs. an f64 direct sum (B0 + τ-dependent quadratic + per-sample $\Delta C + \phi_0$) [measured]:

stencil order P	$L = 6$	$L = 8$	$L = 10$
0 (plain window)	$3.8 \cdot 10^{-2}$	–	–
1	$2.4 \cdot 10^{-3}$	$1.7 \cdot 10^{-3}$	$1.7 \cdot 10^{-3}$
2	$1.8 \cdot 10^{-3}$	$8.5 \cdot 10^{-5}$	$7.7 \cdot 10^{-5}$

Each order has a floor set by the dropped next term, independent of L. Full model with GNL ($L = 20$, forward / adjoint) [measured, CPU emulation]:

B_0	2.89 T	1.5 T	0.55 T
order 2 (degree-2 fit)	$4.8 \cdot 10^{-3}$ / $8.3 \cdot 10^{-3}$	$8.9 \cdot 10^{-3}$ / $1.5 \cdot 10^{-2}$	$3.1 \cdot 10^{-2}$ / $7.0 \cdot 10^{-2}$
order 3 (degree-4 fit)	$1.2 \cdot 10^{-3}$ / $2.2 \cdot 10^{-3}$	$3.0 \cdot 10^{-3}$ / $6.5 \cdot 10^{-3}$	$2.9 \cdot 10^{-2}$ / $7.3 \cdot 10^{-2}$

At 0.55 T, $|\chi_j|$ reaches 1–1.4 rad, which is beyond the Taylor range ($|\chi|^3/6$). That is the regime of construction (b).

(b) Least-squares stencils for large chirps

For $|\chi| \gtrsim 1$ we fit Equation (9) directly on a lattice Ω_h of the FOV, with $\Phi_{\boldsymbol{w}} = \sum_q w_q m_q$:

$$\Gamma(\cdot; \boldsymbol{w}, \boldsymbol{\kappa}) = \operatorname*{arg\,min}_{\operatorname{supp}\Gamma \subset \boldsymbol{\kappa} + [-S,S]^3} \sum_{\boldsymbol{r} \in \Omega_h} \left| \sum_{\boldsymbol{u}} \Gamma(\boldsymbol{u}) e^{-i\boldsymbol{u}\cdot\boldsymbol{r}} - \hat{\varphi}(\boldsymbol{r}) e^{-i\boldsymbol{\kappa}\cdot\boldsymbol{r} - i\Phi_{\boldsymbol{w}}(\boldsymbol{r})} \right|^2 + \varrho \|\Gamma\|^2. \tag{14}$$

Write $\Gamma = \varphi \star \hat{g}(\cdot; \boldsymbol{w})$, where $\hat{g}$ is the compact spectrum of the chirp over the FOV. The family $\{\hat{g}(\cdot; \boldsymbol{w})\}_{\boldsymbol{w} \in \mathcal{W}}$ is compressed by a POD basis $\{B_l\}_{l \le L_p}$ with DEIM points $\{\boldsymbol{r}_i\}$:

$$\hat{g}(\cdot; \boldsymbol{w}) \approx \sum_{l=1}^{L_p} \beta_l(\boldsymbol{w}) B_l, \qquad \boldsymbol{\beta}(\boldsymbol{w}) = \left[B_l(\boldsymbol{r}_i)\right]_{il}^{-1} \left[e^{-i\Phi_{\boldsymbol{w}}(\boldsymbol{r}_i)}\right]_i. \tag{15}$$

Per sample this costs L_p sincos, one $L_p \times L_p$ solve and an $L_p \times (2S+1)^3$ contraction, followed by a separable convolution with φ. The fitted stencil is much narrower than a DFT of a windowed chirp: $S = 3$–4 instead of 8–10 for 10^{-3} at 1 rad. Training on the sequence's own $\boldsymbol{w}_j$ instead of the coefficient box gives $L_p = 31 \to 13$. Taps per axis J at tolerance 10^{-3} (1-D, least-squares stencil with a shared correction function) [measured]:

$\Phi_{\max}$ [rad]	0	1	2	4	8
J at $\sigma = 2$	4	8	10	13	18
J at $\sigma = 1.25$	8	8	10	14	18

At $\sigma = 1.25$ a chirp up to about 1 rad is free, because the plain stencil is already 8 wide. Beyond that $J \approx \max(J_{\text{plain}}, \text{chirp spread})$. In k-space the Gram matrix is banded, $\hat{\mathsf{N}} = \Gamma^{\mathsf{H}}\Gamma$, with bandwidth equal to the stencil support. The diagonal fields (z^2, $x^2{+}y^2$) give an exactly separable $\hat{g}$; with the cross terms the CP rank is 3–4 at 10^{-3}. *Status* [measured, $N = 128$, $K = 10^6$, $C = 8$, $S = 3$]: $A^{\mathsf{H}}A = 18.2$ ms/coil (39 plain-Toeplitz applications) with stored fp16 stencils and a tiled scatter. This is the only construction with one FFT pair per coil independent of Φ and L. It handles the concomitant phase only, with no B0 segments yet.

Choice

construction	FFT pairs/coil	taps/sample	valid for	status
Taylor, $P = 2$	L	W^3	$\lvert\chi\rvert \lesssim 0.7$	production
Taylor, order 3	L	W^3	+ GNL residual	built
least squares	1	$(2S{+}W)^3$	$\Phi \lesssim 8$	built, no B0
Weyl/Toeplitz (Section 1.7)	4–35	0	$\Phi \lesssim 3$	built, unused

Not built: least-squares stencils for the post-split residual χ_j inside the segmented operator. That would combine L B0 segments with $|\chi| > 1$ (low field, strong GNL).

1.4 The full operator

Time-resolved sets add a periodic B-spline in cardiac time, $x(\boldsymbol{r}, t) = \sum_f B_f(t) x_f(\boldsymbol{r})$, linear ($N_f = 2$) or cubic ($N_f = 4$):

$$(Ax)_{cj} = \mathrm{e}^{-\mathrm{i}\phi_{0j}} \sum_{\boldsymbol{u}} \Gamma_j(\boldsymbol{u}) \sum_{f \in \mathcal{F}(t_j)} B_f(t_j) \sum_{\ell=1}^{L} a_\ell(\tau_j)\, G_{cf\ell}(\boldsymbol{u}), \qquad G_{cf\ell} = \mathcal{F}_\sigma\big[\hat{\varphi}^{-1} S_c D_\ell x_f\big]. \tag{16}$$

FFTs scale with the number of frames and gather/scatter with N_f. Several images on the same samples share one tile pass: $y = \sum_k w_k \odot Ax_k$, $x_k = A^{\mathsf{H}}(\bar{w}_k \odot y)$, $k \le 4$ (separate-species fat: $w_{\text{fat}} = c(\tau_j)$). The adjoint accumulates per tile without atomics (exact CSR tile→sample lists). Cost per call, RTX 4090, $180 \times 180 \times 28$, $M = 2.2 \cdot 10^5$, 12 coils [measured, ms]:

operator	forward	adjoint
plain NUFFT, σ=2, W=6	6.0	24.7
segmented, L=1, P=0	5.7	8.7
segmented, L=8, $P = 0/1/2$	30.1 / 30.9 / 31.7	31.2 / 31.8 / 30.1
segmented, L=10/12, P=2	40.4 / 49.2	35.3 / 41.0

The $P = 2$ stencil costs the same as $P = 0$ because Γ_j is shared by all $C \cdot L$ channels. About half of the call is the memory-bound pad/FFT/crop.

1.5 Velocity parametrisation

Model. With encode moments $\boldsymbol{k}_e = \gamma \boldsymbol{M}_{1,e}$ and temporal bases $\beta^z_\kappa, \beta^v_\kappa$ (periodic B-splines, independent of the operator frames),

$$x_e(\boldsymbol{r},t) = z(\boldsymbol{r},t)\,\mathrm{e}^{\mathrm{i}\varphi_e(\boldsymbol{r})}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{k}_e\cdot\boldsymbol{v}(\boldsymbol{r},t)}, \quad z = \sum_{\kappa\le K_z} \beta^z_\kappa(t) z_\kappa, \quad \boldsymbol{v} = \sum_{\kappa\le K_v} \beta^v_\kappa(t)\boldsymbol{v}_\kappa, \quad \varphi_e = \sum_p a_{ep} P_p. \tag{17}$$

Here $z \in \mathbb{C}$ carries the magnitude and every phase common to all encodes, $\boldsymbol{v} \in \mathbb{R}^3$ is the velocity, and P_p are Legendre polynomials (residual eddy-current or model phase, degree ≤ 3). The unknowns $\theta = (z_\kappa, \boldsymbol{v}_\kappa, a)$ are 5 reals per voxel and node instead of $2E$. The model is linear in z, and dual-VENC unwrapping is part of the solve. The operator sees images at frames t_f: $x_{ef} = x_e(\cdot, t_f)$. Interpolation to each sample's t_j happens inside A (Equation (16)). Each (frame, encode) set $s = (f, e)$ has its own A_s, and

$$F(\theta) = \tfrac{1}{2}\sum_s \omega_s \big\| A_s\, x_s(\theta) - y_s \big\|^2 + \tfrac{\lambda}{2}\sum_\kappa \big\| w^{1/2}\boldsymbol{v}_\kappa \big\|^2, \tag{18}$$

where ω_s are the VENC-continuation weights and w a static-tissue mask (robust IRLS fit of the warm-start phase differences, refined by pulsatility).

Gradient. One gradient is one image-space gradient per set followed by a pointwise vector–Jacobian product. With $g_s = A_s^{\mathsf{H}}(A_s x_s - y_s)$ and $u_s = \mathrm{e}^{\mathrm{i}\varphi_e - \mathrm{i}\boldsymbol{k}_e\cdot\boldsymbol{v}(t_f)}$ (so $x_s = z(t_f)u_s$),

$$\begin{aligned} \nabla_{z_\kappa} F &= \sum_s \omega_s \beta^z_\kappa(t_f)\,\overline{u_s}\, g_s, \qquad \nabla_{a_{ep}} F = -\sum_{s\ni e}\omega_s \sum_{\boldsymbol{r}} P_p\, \mathrm{Im}\big(\overline{g_s}\, x_s\big), \\ \nabla_{\boldsymbol{v}_\kappa} F &= \sum_s \omega_s \beta^v_\kappa(t_f)\,\boldsymbol{k}_e\, \mathrm{Im}\big(\overline{g_s}\, x_s\big) + \lambda w \boldsymbol{v}_\kappa. \end{aligned} \tag{19}$$

Proof: $\mathrm{d}x_s = u_s\,\mathrm{d}z - \mathrm{i}(\boldsymbol{k}_e\cdot\mathrm{d}\boldsymbol{v})x_s + \mathrm{i}P_p x_s\,\mathrm{d}a_{ep}$ and $\mathrm{d}F = \mathrm{Re}\langle g_s, \mathrm{d}x_s\rangle$. The cost is that of the complex model (one forward and one adjoint per set) plus $O(|\Omega|K)$ pointwise work. It does not grow with K_z, K_v.

Metric. Gauss–Newton $H = J^{\mathsf{H}}\mathsf{N}J$ with $\mathsf{N} \approx \hat{n} I$, where $\hat{n} = \mathbb{E}_\xi\langle \xi, \mathsf{N}\xi\rangle / \|\xi\|^2$ per unit coil energy (random probe), gives pointwise blocks per node:

$$M_z^{-1} = \hat{n}\sum_s \omega_s \beta^z_\kappa(t_f)^2, \qquad M_{\boldsymbol{v}}^{-1}(\boldsymbol{r}) = \hat{n}\,|\bar{z}(\boldsymbol{r})|^2 \sum_s \omega_s \beta^v_\kappa(t_f)^2\, \boldsymbol{k}_e\boldsymbol{k}_e^{\top} + \lambda w(\boldsymbol{r}) I + \mu I. \tag{20}$$

These are refreshed every 5 gradients and used as the base metric of ZMSHR1 (Section 1.8). The cross term $\partial^2 F/\partial z \partial \boldsymbol{v} \propto |z|^2 \sin\phi_c \sum_e \boldsymbol{k}_e$ (with ϕ_c an encode-common phase error) is dropped by the block metric. For referenced and dual-VENC encodings $\sum_e \boldsymbol{k}_e \neq 0$, so the solve starts with n z-only iterations, which form a linear subproblem.

Exact phases on the sample side. A degree-1 residual phase is a k-space shift: $A_{\boldsymbol{k}}(x\,\mathrm{e}^{\mathrm{i}(c+\boldsymbol{g}\cdot\boldsymbol{\rho})}) = \mathrm{e}^{\mathrm{i}c} A_{\boldsymbol{k}-\boldsymbol{g}}(x)$, verified to $5\cdot 10^{-7}$. Injected or estimated linear phases therefore cost nothing. GNL: reconstruct at apparent positions $\boldsymbol{\rho} = u(\boldsymbol{r})$ and correct $\boldsymbol{v}_{\text{true}} = J_u^{-1}\boldsymbol{v}_{\text{app}}$.

Readout motion (not yet in the reconstruction). During the readout the first moment adds $\mathrm{e}^{-\mathrm{i}\gamma \boldsymbol{m}_{1,j}(\tau)\cdot \boldsymbol{v}(\boldsymbol{r})}$. The simulator includes it, so it is a model mismatch of the current reconstruction. It has the form of $\omega(\boldsymbol{r})\tau$. With $\boldsymbol{m}_{1,j} = \bar{\boldsymbol{m}}_1(\tau) + \delta\boldsymbol{m}_{1,j}(\tau)$ (mean over interleaves plus a rotation-dependent part), the planned treatment is

$$\psi(\boldsymbol{r},\tau) \;\leftarrow\; \psi(\boldsymbol{r},\tau) + \gamma\, \bar{\boldsymbol{m}}_1(\tau)\cdot \boldsymbol{v}^{(k)}(\boldsymbol{r}), \tag{21}$$

which is lagged in the iterate $\boldsymbol{v}^{(k)}$, uses the same segments, and refreshes $D_\ell, \boldsymbol{a}$ when $\|\boldsymbol{v} - \boldsymbol{v}^{(k)}\|$ grows. The rotation-dependent part $\delta\boldsymbol{m}_{1,j}\cdot\boldsymbol{v}(\boldsymbol{r})$ is not polynomial in $\boldsymbol{r}$ and is not modelled.

Results (R4, 7-point dual-VENC 1.0/3.5 m/s, 60 s, $R \approx 8$ per frame, 30 iterations, no prior) [measured]:

model	velocity RMSE (m/s)	wrapped
complex images + locally-low-rank prior	0.183	0.4 %
Equation (17), $K_z = K_v = 8$ cubic, VENC continuation	0.138	0.11 %
+ grid 1.25× finer (1.6 mm)	0.130	–
+ injected residual phase 0.6 rad, robust init + anchor	0.166	–
4-point, full physics + GNL, 10 z-only + 30 joint	0.232	0.01 %

1.6 Evaluating the gradient $A^{\mathsf{H}}(Ax - y)$ fast

Cost model. The gradient is $\nabla f = \sum_s A_s^{\mathsf{H}}(A_s x_s - y_s)$, computed as forward, residual, adjoint. With $|\Omega_\sigma| = \sigma^3|\Omega|$ grid points and M_s samples,

$$\begin{aligned} T_\nabla \approx \sum_s \Big[& \underbrace{2CF_sL\left(T_{\mathrm{FFT}}(|\Omega_\sigma|) + T_{\mathrm{pad/crop}}\right)}_{\text{grid side} \propto CF_sL|\Omega_\sigma|} + \underbrace{2CL\,N_f M_s W^3 t_{\mathrm{mac}}}_{\text{gather/scatter}} \\ & + \underbrace{N_f M_s W^3\, n_\gamma t_{\mathrm{mac}}}_{\text{stencil, coil- and segment-free}} + \underbrace{3CM_s\, t_{\mathrm{mem}}}_{\text{residual}} \Big]. \end{aligned} \tag{22}$$

Here F_s is the number of frame grids that set s touches (1 for a static set), and $n_\gamma \le 35$.

Implemented levers.

1. *Shared stencils.* The n_γ term is divided over CLN_f channels, so $P = 2$ costs the same as $P = 0$ (31.7 vs. 30.1 ms).
2. *Small grid.* Bounded window derivatives allow $\sigma = 1.25$: $|\Omega_\sigma| = 0.24\times$ the $\sigma = 2$ grid.
3. *Fused grid passes.* The grid is channel-interleaved [point][$c\cdot L + \ell$] with one partial-axis FFT over all channels. The pad pass forms $\hat{\varphi}^{-1} S_c D_\ell x_f$ for all (c,ℓ) in one pass; the crop pass forms $\sum_c \overline{S_c} \sum_\ell \overline{D_\ell}(\cdot)$ in one multi-input pass. Launch shapes are planned within 48 KB of shared memory and autotuned per set.
4. *Coil compression.* With the whitened Gram $\sum_n y_n y_n^{\mathsf{H}} = U\Sigma^2 U^{\mathsf{H}}$, keep $C_v = \min\{n : \sum_{i\le n} s_i^2 \ge 0.999 \sum_i s_i^2\}$ virtual coils $U_{:,1:C_v}^{\mathsf{H}} y$. Rows are orthonormal, so the noise stays white. On the 32-coil array: $C_v = 16$ and 29 [measured]. The Gram is taken over the Cartesian calibration region: a centre-dense trajectory would weight it towards $k = 0$ and keep too few coils.

5. *Stochastic coil subsets.* With energy-ordered virtual coils, keep the leading set $\mathcal{K}$ and draw $\mathcal{D}_k$ from the remaining ones:

$$\tilde{\nabla} f = \sum_{c \in \mathcal{K}} g_c + p \sum_{c \in \mathcal{D}_k} g_c, \qquad \mathbb{E}\, \tilde{\nabla} f = \nabla f, \tag{23}$$

where g_c is the gradient of coil c and p is the inverse inclusion probability. With random reshuffling the epoch mean is exact. The step uses $\alpha = 1/\max\big(\lambda_{\max}(\mathsf{N}), \max_{\text{draws}} \lambda_{\max}(\mathsf{N}_{\mathcal{K}} + p\mathsf{N}_{\mathcal{D}})\big)$. A subset is held for $h \geq 2$ iterations, so the ZMSHR1 secant pair satisfies $u = \tilde{\mathsf{N}} s$ exactly. Measured on 8 virtual coils: the epoch mean equals the full gradient to $7 \cdot 10^{-6}$; $\lambda_{\max}$ inflation is 1.42 when drawing 4 of 8 and 0.90 when keeping 2 and drawing 3 of 6. At 60 gradient-equivalents PnP-ISTA has 10–16 % lower error; ZMSHR1 does not gain.

6. *Shared gridding* of $k \leq 4$ images on the same samples (Section 1.4). One-set water+fat segments use L channels instead of $2L$ (Section 1.2).

7. *Streaming.* The k-space data y_s come from a host store (dense, lz4/zstd blocks, or file). Sample tables are kept packed (zstd, ×6.2) and decoded into pinned staging buffers. A worker decodes set $s{+}1$ while the GPU runs set s, so the wall time per set is $\max(T_{\text{GPU}}, T_{\text{decode}})$ rather than their sum. A coil subset reads only its own rows.

8. *Lagged control variate with a reduced-segment surrogate.* The surrogate $\tilde{E}$ is the same operator with $L_{\text{s}} < L$ segments on the same sample tables, and $\tilde{\mathsf{N}} = \tilde{E}^{\mathsf{H}}\tilde{E}$:

$$\begin{aligned} \tilde{g}_k &= \tilde{\mathsf{N}}(x_k - x_a) + \nabla f(x_a), \qquad x_a \leftarrow x_k \text{ if } \|x_k - x_a\| > \tau \|x_k\|, \\ \tilde{f}_k &= f(x_a) + \operatorname{Re}\langle \nabla f(x_a), x_k - x_a \rangle + \tfrac{1}{2}\|\tilde{E}(x_k - x_a)\|^2. \end{aligned} \tag{24}$$

The error is $(\tilde{\mathsf{N}} - \mathsf{N})(x_k - x_a)$, which vanishes at the fixed point. Cheap steps read no data and cost $\approx L_{\text{s}}/L$ of an exact step. A secant pair is kept only within one anchor (or between two exact gradients); otherwise the metric step is used. L_{s} is chosen as the smallest value with $\|(E - \tilde{E})x\| \leq 0.03\|Ex\|$ on the warm start. Measured (test operator, 60 ZMSHR1 iterations, $\tau = 0.1$): with a mismatch of 2.4 % (1.5 phase cycles, $L_{\text{s}} = 3$ of 8) or 15 % at 0.3 cycles, 7 exact + 53 cheap gradients reach the exact run's error. With 7–8 % at ≥ 1.5 cycles it needs 33–45 exact gradients and ends 2–3× further away. On sim_point (4.3 cycles, $L = 10$, chosen $L_{\text{s}} = 4$ at 2.6 %), 30 iterations took 10 exact + 20 cheap gradients and 28 % less time, at a 35 % higher final f.

9. *Pruned grid FFT.* The pad pass places the image at $(a - N/2) \bmod \sigma N$ per axis, so only a wrapped block $[0, \lceil N/2 \rceil) \cup [\sigma N - \lfloor N/2 \rfloor, \sigma N)$ is non-zero. The forward reads only that block: the x pass runs over lines with (y, z) in the block, the y pass over planes with z in the block, and the z pass over all lines. Nothing outside is read, so the grid is never zeroed. The inverse mirrors it and writes only the block that the crop reads. Traffic per direction is $\approx \frac{1}{3}(h_y h_z + h_z + 1) \cdot \frac{1}{2}(1 + h)$ with $h = 1/\sigma$. Measured (FFT alone, $180 \times 180 \times 50$, 16 channels): forward including the former memset 2.01 → 0.98 ms, inverse 1.35 → 0.96 ms; at $\sigma = 2$ both 2.8× faster. Results equal the full transform to $3 \cdot 10^{-7}$.

10. *Fused residual.* The gather accumulates $y \mathrel{+}= \Gamma G$, so the accumulator starts at $-y_s$ and the result is $r_s = A_s x_s - y_s$. There is no subtraction pass and no persistent residual buffer: one set's residual lives in a stream-ordered temporary instead of a second copy of all data.

Not yet implemented.

1. *Segment Gram (Toeplitz) form* [4]:

$$\mathsf{N} = \sum_c \overline{S_c} \sum_{\ell\ell'} \overline{D_\ell}\, \mathsf{T}[g_{\ell\ell'}]\, D_{\ell'} S_c, \qquad g_{\ell\ell'}(\boldsymbol{\Delta}) = \sum_j \overline{a_\ell(\tau_j)} a_{\ell'}(\tau_j) \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{\Delta}} \quad \text{(no per-sample chirp)}.$$

   It needs $L(L+1)/2$ kernels of $(2N)^3$ per set, which does not fit for time-resolved sets (≈ 80 sets). It is useful for few-set problems.

1.7 Algebra of the normal operator

Coil maps are pointwise multiplications, so $\mathsf{N} = \sum_c \overline{S_c}\, \mathsf{N}_0\, S_c$ with N_0 coil-free. Below, $\mathsf{N} := \mathsf{N}_0$, $\Phi_j = \Theta_j - \boldsymbol{k}_j \cdot \boldsymbol{r}$, and X_a, $\mathsf{P} = -\mathrm{i}\nabla$, $\mathsf{T}[g]\rho = g * \rho$:

$$\mathsf{N}(\boldsymbol{r}, \boldsymbol{r}') = \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot(\boldsymbol{r}-\boldsymbol{r}')} \mathrm{e}^{\mathrm{i}(\Phi_j(\boldsymbol{r})-\Phi_j(\boldsymbol{r}'))}, \qquad \mathsf{N} = \sum_j \mathsf{Q}_j\, \mathsf{T}[\mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{\Delta}}]\, \mathsf{Q}_j^{\mathsf{H}}, \quad \mathsf{Q}_j = \mathrm{e}^{\mathrm{i}\Phi_j(\mathsf{X})}. \tag{25}$$

Lemma 3. $[\mathsf{X}_a, \mathsf{T}[g]] = \mathsf{T}[\Delta_a g]$, *hence* $\mathrm{ad}_{\mathsf{X}}^{\beta}\mathsf{T}[g] = \mathsf{T}[\boldsymbol{\Delta}^\beta g]$. *(Checked to* $6 \cdot 10^{-14}$*.)*

Lemma 4 (midpoint)**.** *For* $\Phi = \frac{1}{2}\boldsymbol{r}^\top M \boldsymbol{r} + \boldsymbol{b}\cdot\boldsymbol{r} + c$, $M = M^\top$, $\bar{\boldsymbol{r}} = \frac{1}{2}(\boldsymbol{r}+\boldsymbol{r}')$, $\boldsymbol{\Delta} = \boldsymbol{r} - \boldsymbol{r}'$: $\Phi(\boldsymbol{r}) - \Phi(\boldsymbol{r}') = (M\bar{\boldsymbol{r}} + \boldsymbol{b})\cdot\boldsymbol{\Delta}$ *exactly.*

Proposition 5 (exact Weyl symbol)**.** *For quadratic* Φ_j *(here* $M_j = \sum_q w_{jq}\nabla^2 m_q$*),*

$$\sigma_{\mathsf{N}}(\bar{\boldsymbol{r}}, \boldsymbol{k}) = \sum_j \delta\big(\boldsymbol{k} - \boldsymbol{k}_j - M_j\bar{\boldsymbol{r}} - \boldsymbol{b}_j\big), \qquad \mathsf{N} = \sum_j \delta(\Pi_j - \boldsymbol{k}_j), \quad \Pi_j = \mathsf{Q}_j\mathsf{P}\mathsf{Q}_j^{\mathsf{H}} = \mathsf{P} - M_j\mathsf{X} - \boldsymbol{b}_j, \tag{26}$$

with commuting components ($[\Pi_{j,a}, \Pi_{j,b}] = \mathrm{i}(M_{j,ba} - M_{j,ab}) = 0$*) and no higher-order corrections: conjugation by a quadratic phase is metaplectic, i.e. the shear* $(\boldsymbol{r}, \boldsymbol{k}) \mapsto (\boldsymbol{r}, \boldsymbol{k} - \nabla\Phi)$ *[7]. The effective trajectory at* $\bar{\boldsymbol{r}}$ *is* $\boldsymbol{k}_j + M_j\bar{\boldsymbol{r}} + \boldsymbol{b}_j$*.*

Proposition 6 (commutator and Δ-forms)**.** *With* $g^\alpha = \sum_j w_j^\alpha \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{\Delta}}$*,*

$$\begin{aligned} \mathsf{N} &= \sum_\alpha \frac{\mathrm{i}^{|\alpha|}}{\alpha!} \mathrm{ad}_{m_1}^{\alpha_1} \cdots \mathrm{ad}_{m_4}^{\alpha_4} \mathsf{T}[g^\alpha], \\ \mathsf{N} = \sum_n \mathsf{K}_n, \qquad K_n(\boldsymbol{r}, \boldsymbol{r}') &= \frac{\mathrm{i}^n}{n!} \sum_j \big[(M_j\bar{\boldsymbol{r}} + \boldsymbol{b}_j)\cdot\boldsymbol{\Delta}\big]^n \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j\cdot\boldsymbol{\Delta}} = \sum_{|\beta|\le n} \bar{\boldsymbol{r}}^\beta g_{n\beta}(\boldsymbol{\Delta}). \end{aligned} \tag{27}$$

The kernel $\bar{\boldsymbol{r}}^\beta g(\boldsymbol{\Delta})$ *is the Weyl-symmetrised product* $2^{-|\beta|}\sum_{\gamma\le\beta}\binom{\beta}{\gamma}\mathsf{X}^\gamma\mathsf{T}[g]\mathsf{X}^{\beta-\gamma}$*. Every truncation is exactly Hermitian. Order* n *costs* $\binom{n+3}{3}$ *FFT pairs (4, 10, 20, 35), and the remainder is* $\le \sum_j (2\Phi_{\max})^{n+1}/(n+1)!$*.*

Proofs: Hadamard $\mathrm{e}^{\mathrm{i}A}T\mathrm{e}^{-\mathrm{i}A} = \sum_n \frac{\mathrm{i}^n}{n!}\mathrm{ad}_A^n T$ with commuting $m_q(\mathsf{X})$ and the multinomial theorem; for the Δ-form expand in the pair phase of Lemma 4. (Order 8 reproduces the dense N to $5\cdot10^{-12}$.) First order: $\mathrm{i}\sum_q[m_q, \mathsf{T}[g^{e_q}]] = \frac{\mathrm{i}}{2}\sum_{q,a}\{\partial_a m_q(\mathsf{X}), \mathsf{T}[\Delta_a g^{e_q}]\}$ needs 7 kernels and 4 FFT pairs.

Segmented Toeplitz form with derivative multipliers. For one segment, $T_\ell = \mathsf{T}[\sum_n a_{\ell n} \mathrm{e}^{\mathrm{i}\boldsymbol{k}_n \cdot \boldsymbol{\Delta}}]$ with multiplier τ_ℓ, and a quadratic residual $C_n \approx c(\boldsymbol{k}_n)$:

$$\mathsf{N}_\ell \approx T_\ell + \mathrm{i}\sum_{pq}\{\mathsf{X}_p, [\mathsf{X}_q, S_{\ell,pq}]\} - \tfrac{1}{2}\sum_{pqst}\{\mathsf{X}_p, \{\mathsf{X}_s, [\mathsf{X}_q, [\mathsf{X}_t, S_{\ell,pq,st}]]\}\}, \qquad \hat{S}_{\ell,pq} = c_{pq}\tau_\ell,\ \hat{S}_{\ell,pq,st} = c_{pq}c_{st}\tau_\ell, \tag{28}$$

$$\mathcal{F}\{\mathrm{i}\{\mathsf{X}_p, [\mathsf{X}_q, S]\}v\} = -\mathrm{i}\big[(\partial_{pq}\hat{S})V + 2(\partial_q \hat{S})(\partial_p V)\big]. \tag{29}$$

The transform count stays $2L$. However, τ_ℓ fills the whole $2N$ period and cannot be differenced, so it needs 35 stored derivative kernels per segment. $c(\boldsymbol{k})$ (degree 2 in $\boldsymbol{k}$) still leaves 0.066 rad. Not used for time-resolved sets.

Operator error vs. FFT pairs (8^3, 960 samples, rotated spirals; $\|\mathsf{N} - \mathsf{N}_{\text{approx}}\|_F / \|\mathsf{N} - \mathsf{N}^0\|_F$) [measured]:

method (FFT pairs)	$\Phi_{\max}$=0.3	1	3
Taylor of $\mathrm{e}^{-\mathrm{i}\Phi}$, P=3 (35)	$4.5 \cdot 10^{-4}$	$1.6 \cdot 10^{-2}$	0.37
Weyl n=1 (4)	$6.2 \cdot 10^{-2}$	0.21	0.61
Weyl n=2 (10)	$3.5 \cdot 10^{-3}$	$3.8 \cdot 10^{-2}$	0.34
Weyl n=4 (35)	$7.2 \cdot 10^{-6}$	$8.9 \cdot 10^{-4}$	$7.3 \cdot 10^{-2}$
SVD–Tucker L=12 (12)	$5.9 \cdot 10^{-4}$	$2.1 \cdot 10^{-3}$	$9.8 \cdot 10^{-3}$

$\|\mathsf{N} - \mathsf{N}^0\| / \|\mathsf{N}\| = 0.8, 2.8, 7.9\,\%$. No expansion has an FFT count independent of Φ: the Slepian number $\approx (\Phi_{\max}/\pi)c$ sets a floor of order L. Subdividing the samples (butterfly) increases the total basis count ($42 \to 51/80/132$ for 2/4/8 time bins at $\Phi \approx 9$).

1.8 Preconditioned plug-and-play iteration

The unknowns are one image per set $t = (f, e)$ (frame f = cardiac × respiratory bin, encode e), and the data term is separable over the sets:

$$f(x) = \tfrac{1}{2}\sum_t \|E_t x_t - y_t\|_2^2, \qquad \nabla_t f = \mathsf{N}_t x_t - E_t^{\mathsf{H}} y_t, \qquad \mathsf{N}_t = E_t^{\mathsf{H}} E_t, \qquad \nabla^2 f = \operatorname{blockdiag}_t \mathsf{N}_t. \tag{30}$$

With whitened coils, f is the negative Gaussian log-likelihood up to a constant (no density weights). The iteration (P^{2}nP [13]) is

$$\begin{aligned} x_{k+1} &= \mathcal{D}\big(x_k - \alpha\, \mathsf{P}_k g_k\big), \qquad g_k = \nabla f(x_k), \\ \mathsf{P}_k &= \mathrm{ZMSHR1}\big(M;\, s_k = x_k - x_{k-1},\, u_k = \alpha(g_k - g_{k-1})\big), \end{aligned} \tag{31}$$

with one operator application per iteration and a rank-one update of a base metric M. The step is $\alpha = 1/\lambda_{\max}(M\mathsf{N})$, from a power iteration. On our trajectories the Fourier diagonal of N spans 10^3–10^6 from DC to Nyquist. The base metric replaces density compensation without entering the data term. The velocity model uses the pointwise Gauss–Newton blocks of Section 1.5 instead.

The circulant metric

Take the plain (physics-free) model of a set with samples $\boldsymbol{k}_j$ (radians per voxel) on the grid $N = N_1 \times N_2 \times N_3$ (n_v voxels), and let F be the unitary N-point DFT. The coil-free normal operator is Toeplitz,

$$T_t(\boldsymbol{r}, \boldsymbol{r}') = h_t(\boldsymbol{r} - \boldsymbol{r}'), \qquad h_t(\boldsymbol{n}) = \sum_{j \in t} \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j \cdot \boldsymbol{n}}, \qquad \boldsymbol{n} \in \{-N_a{+}1, \ldots, N_a{-}1\}^3.$$

Its optimal circulant approximation in Frobenius norm (T. Chan) folds the triangle-windowed kernel onto the N grid,

$$c_t(\boldsymbol{n} \bmod N) = \sum_{\boldsymbol{n}} \prod_a \Big(1 - \frac{|n_a|}{N_a}\Big) h_t(\boldsymbol{n}), \qquad \rho_t = \sqrt{n_v}\, F c_t = \sum_{j \in t} \prod_a K_{N_a}\big(\kappa_a - \tfrac{N_a}{2\pi} k_{j,a}\big) \;\geq 0,$$

with K_{N_a} the Fejér kernel. So ρ_t is the sample density on the grid frequencies $\boldsymbol{\kappa}$, smoothed over about one bin. With coil maps, $\mathsf{N}_t = \sum_c S_c^{\mathsf{H}} T_t S_c$. Replacing T_t by $C_t = F^{\mathsf{H}} \operatorname{diag}(\rho_t) F$ and using $\langle e_{\boldsymbol{\kappa}'}, S_c e_{\boldsymbol{\kappa}} \rangle = n_v^{-1/2} \hat{S}_c(\boldsymbol{\kappa}' - \boldsymbol{\kappa})$, with $\hat{S}_c = \sqrt{n_v} F S_c$, gives the Fourier diagonal

$$\lambda_t(\boldsymbol{\kappa}) = \langle e_{\boldsymbol{\kappa}}, \mathsf{N}_t e_{\boldsymbol{\kappa}} \rangle \approx \sum_{\boldsymbol{\kappa}'} \rho_t(\boldsymbol{\kappa}')\, P(\boldsymbol{\kappa}' - \boldsymbol{\kappa}), \qquad P(\boldsymbol{\kappa}) = \frac{1}{n_v} \sum_c |\hat{S}_c(\boldsymbol{\kappa})|^2. \tag{32}$$

This is the density convolved with the coil power spectrum. Inverting ρ_t alone over-amplifies the modes next to the DC spike, which the maps smear onto their neighbours: $\lambda_{\max}(M\mathsf{N})$ went from 38 to 9 on the spiral. Two floors complete it:

- in-plane corners $|\boldsymbol{\kappa}_{xy}| > 0.95$ (no samples) are raised to the median of the ring 0.85–0.95;
- $\lambda_\varepsilon = \max(\lambda, \varepsilon \lambda_{\max})$ with $\varepsilon = 10^{-3}$.

With $D = \sum_c |S_c|^2 + 10^{-3} \max \sum_c |S_c|^2$ and $\bar{D}$ the mean of $\sum_c |S_c|^2$ (already contained in λ),

$$M = D^{-1/2} F^{\mathsf{H}} \operatorname{diag}\big(\bar{D}/\lambda_\varepsilon\big) F D^{-1/2}, \qquad M^{-1} = D^{1/2} F^{\mathsf{H}} \operatorname{diag}\big(\lambda_\varepsilon/\bar{D}\big) F D^{1/2} \approx \mathsf{N}_t. \tag{33}$$

M is Hermitian positive definite. It does not depend on y; it depends only on the nominal trajectory and the maps. One application costs two N-point FFTs and three pointwise products. At $254 \times 144 \times 114$: 0.77 ms with the arrays on the device, 4.26 ms host-resident (two N-grid arrays uploaded per application). Building it takes 2.0 s from the $3.0 \cdot 10^7$ samples of one encode (Toeplitz kernel on the $2N$ grid, fold, coil spectra) [measured]. On the ph0001 yarn trajectory (encode 0 pooled), $\lambda/\lambda_{\max}$ has median $4 \cdot 10^{-2}$ at 0.05, $5 \cdot 10^{-3}$ at 0.2 and $2 \cdot 10^{-3}$ at Nyquist. The 10^{-3} floor clips no mode inside Nyquist [measured].

Sharing between encodes and frames

The default is one metric per encode, built from the union of the samples of all frames of that encode: $M_t := M_{e(t)}$. A per-set metric costs 133 MB of host memory at 2 mm, 37 GB for 280 sets.

- *Encodes.* Readout i plays rotation $\lfloor i/7 \rfloor$ with encode i mod 7, and pooling keeps every readout. The pooled densities of the seven encodes are therefore identical: their ratio is 1.000 in every $|\boldsymbol{\kappa}|$ shell [measured]. The seven M_e coincide, and one metric would suffice.
- *Frames.* A set holds $n_t/n_e \approx 1/40$ of its encode's samples (20 cardiac × 2 respiratory bins). The rotations reach a frame through the ECG timing, quasi-uniformly. Hence $\mathbb{E}\rho_t \approx (n_t/n_e)\rho_e$ and
$$M_t^{\text{own}} \approx \frac{n_e}{n_t} M_e \quad \text{in expectation.}$$
The shared metric is therefore right up to a scalar, which τ_t below absorbs (and α through $\lambda_{\max}$).
- *Per-set deviation.* We measured $\rho_t/((n_t/n_e)\rho_e)$ for one cardiac frame, both respiratory bins pooled (sample histograms on the N grid, $\sigma = 0.5$ bin) [measured]:

– median 1.00 (p10–p90: 0.76–1.23) below 0.05 Nyquist;
– 0.96 (0.45–1.6) at 0.05–0.1;
– 0.62–0.68 (0.07–2.4) beyond 0.2.

The skewed spread is angular undersampling: 380 of 7592 rotations. A reconstruction set is 2× sparser still.

- *What no circulant captures.* Above ≈ 0.1 Nyquist a single set's Fourier diagonal fluctuates by an order of magnitude from bin to bin. Its normal operator also has strong off-diagonal (aliasing) structure, which no circulant can represent. A per-set circulant would track the diagonal but not the coupling. The shared M_e captures what all sets have in common: the radial density, i.e. the 10^3 range from DC to Nyquist. The rest is left to the rank-one term and to the prior.

- *Physics.* M is built for the plain model. The full model's phases (B0, fat, concomitant fields, GNL) are unimodular per sample: they leave $\operatorname{tr} \mathsf{N}_t$ unchanged and spread the Fourier diagonal over the chirp bandwidth. For the full model at 2 mm with joint water–fat, $\lambda_{\max}(M\mathsf{N}) = 11.2$ [measured].

ZMSHR1 in the metric

Change variables by $x = M^{1/2}\hat{x}$. Then $\hat{s} = M^{-1/2}s$, and the gradient difference transforms covariantly, $\hat{u} = M^{1/2}u$. ZMSHR1 [13] on the pair $(\hat{s}, \hat{u})$ proceeds as follows.

- *Mixing.* $\hat{v} = a\hat{s} + (1-a)\hat{u}$, where a is the smallest value in $\{0, \frac{1}{2000}, \dots, 1\}$ with $\langle \hat{s}, \hat{v}\rangle \geq \theta_1 \|\hat{s}\|^2$ and $\|\hat{v}\|^2 \leq \theta_2 \langle \hat{s}, \hat{v}\rangle$ ($\theta_1 = 2 \cdot 10^{-6}$, $\theta_2 = 200$).

- *Scale.* $\tau = \beta - \left(\beta^2 - \|\hat{s}\|^2/\|\hat{v}\|^2\right)_+^{1/2}$ with $\beta = \|\hat{s}\|^2/\langle \hat{s}, \hat{v}\rangle$.

- *Update.* $\hat{B} = \tau I + \hat{w}\hat{w}^{\mathsf{H}}/\langle \hat{w}, \hat{v}\rangle$ with $\hat{w} = \hat{s} - \tau\hat{v}$. The rank-one term is dropped if $\langle \hat{w}, \hat{v}\rangle \leq 10^{-8}\|\hat{w}\|\|\hat{v}\|$.

All scalars come from three inner products in the original variables:

$$\|\hat{s}\|^2 = \langle s, M^{-1}s\rangle, \qquad \langle \hat{s}, \hat{u}\rangle = \langle s, u\rangle, \qquad \|\hat{u}\|^2 = \langle u, Mu\rangle.$$

Transforming back gives

$$\begin{gathered} \mathsf{P} = M^{1/2}\hat{B}M^{1/2} = \tau M + \frac{ww^{\mathsf{H}}}{\langle w, z\rangle}, \\ w = M^{1/2}\hat{w} = (1 - \tau a)s - \tau(1-a)Mu, \qquad z = M^{-1/2}\hat{v} = aM^{-1}s + (1-a)u, \end{gathered} \tag{34}$$

since $\langle \hat{w}, \hat{v}\rangle = \langle w, z\rangle$. The step $\mathsf{P}g = \tau Mg + (\langle w, g\rangle/\langle w, z\rangle)\, w$ needs $M^{-1}s$, Mu and Mg: three metric applications. The code recomputes the secant in a second pass and uses five. $M = I$ recovers plain ZMSHR1. The first iteration, and any iteration after a change of coil subset, takes the metric step $\mathsf{P} = M$.

The variants differ in how much structure the quasi-Newton matrix shares:

- `zmshr1m`: one triple (a, τ, c) from inner products summed over all sets, i.e. one rank-one term over the whole stack;

- `zmshr1ms`: one triple per set,

$$\mathsf{P} = \mathrm{blockdiag}_t\Big(\tau_t M_{e(t)} + \frac{w_t w_t^{\mathsf{H}}}{\langle w_t, z_t\rangle}\Big),$$

  i.e. one scalar and one rank-one term per (frame, encode). Since $\nabla^2 f$ is block diagonal (30), this is the matching structure. A set with a vanishing secant falls back to $(a, \tau) = (1, 1)$.

- `zmshr1s`: the same per-set form with $M = I$, i.e. *no* Fourier metric.

For 280 sets at 2 mm the metric costs $280 \times 5 \times 4.26\,\mathrm{ms} \approx 6\,\mathrm{s}$ per iteration (1.1 s device-resident). Per iteration the gradient takes 430 s with the whole out-and-back readout and 234 s with its outgoing half; the LLR prior takes 15 s on one GPU (f64 Gram matrices and batched eigendecompositions, blocks split over the GPUs; 125 s on the CPU) [measured], so the metric adds 1.5–3 %.

What the metric does to the estimator

Proposition 7. *Let* P *be fixed and Hermitian positive definite.*

(i) Without a prior ($\mathcal{D} = \mathrm{id}$*), convergent iterations reach a minimiser of* f*, i.e. a maximum-likelihood estimate. If* E *has a null space, they reach the minimiser closest to* x_0 *in* $\|\cdot\|_{\mathsf{P}^{-1}}$*.*

(ii) With a fixed-strength proximal denoiser $\mathcal{D} = \mathrm{prox}_{\theta R}$ *(Euclidean; our LLR singular-value threshold,* $\theta = \kappa\hat{\sigma}$*), every fixed point satisfies*

$$0 \in \nabla f(x^\star) + \frac{\theta}{\alpha}\,\mathsf{P}^{-1}\partial R(x^\star). \tag{35}$$

Proof. (i) For constant P the iterates satisfy $x_k - x_0 \in \mathsf{P}\,\mathrm{range}\,E^{\mathsf{H}}$. Among the minimisers, x_∞ is the one with P^{-1}-orthogonal correction; for PCG this is classical. (ii) $x^\star = \mathrm{prox}_{\theta R}(z)$ if and only if $z - x^\star \in \theta\,\partial R(x^\star)$. Put $z = x^\star - \alpha\mathsf{P}\nabla f(x^\star)$. □

Consequences:

- *No prior.* The warm starts (pooled PCG, $\kappa = 0$) are maximum-likelihood with or without the metric; the metric only changes how fast they converge. A Fourier-diagonal M hardly mixes the unsampled modes (the corners of the $\boldsymbol{\kappa}$ cube) into the sampled ones, so in practice the estimate equals that of $\mathsf{P} = I$.

- *Per-set scalar metric* (`zmshr1s`, $\mathsf{P} = \mathrm{blockdiag}(\tau_t I)$): (35) is the stationarity condition of f plus R with a per-set weight $\theta/(\alpha\tau_t)$. For an LLR block coupling sets of different τ_t, this means a column-weighted nuclear norm.

- *Circulant metric* ($\mathsf{P} \approx \tau M$, $M^{-1} \approx \mathsf{N}$): the prior's subgradient is filtered by the data's curvature. $\mathsf{N}\partial R$ is in general not the subdifferential of any function, so the fixed point is a well-defined estimator but not a MAP estimate.

- *Local quadratic picture.* Take $R \approx \frac{\gamma}{2}\|x\|^2$ on the tangent space at $x^\star$ and $\mathsf{N} \approx \mathrm{diag}(n(\boldsymbol{\kappa}))$ in Fourier. Mode by mode,

$$\hat{x}^\star(\boldsymbol{\kappa}) = \frac{n(\boldsymbol{\kappa})}{n(\boldsymbol{\kappa}) + \theta\gamma/(\alpha\tau)}\,\hat{x}_{\mathrm{ML}}(\boldsymbol{\kappa}) \quad (\mathsf{P} = \tau I), \qquad \hat{x}^\star(\boldsymbol{\kappa}) = \frac{1}{1 + \theta\gamma/(\alpha\tau)}\,\hat{x}_{\mathrm{ML}}(\boldsymbol{\kappa}) \quad (\mathsf{P} = \tau\mathsf{N}^{-1}).$$

The Euclidean metric shrinks the low-density, high-$|\boldsymbol{\kappa}|$ modes most. Even when fully converged it low-passes the image, and partial volume with static tissue then shrinks the lumen velocity. The circulant metric shrinks every mode by the same relative factor. It costs no resolution, but it gives no extra noise suppression at high $|\boldsymbol{\kappa}|$, where the noise variance per mode is $\propto 1/n$.

- *Two effects in the measured shrink.* The shrink of the $\kappa = 1$ runs mixes this estimator effect with non-convergence; separating them needs a $\kappa = 0$ run with `zmshr1ms` [CHECK].

- *Keeping a metric-independent penalised-likelihood estimator* (without ADMM):

  (a) Take the prox in the metric, $x_{k+1} = \operatorname{prox}^{\mathsf{P}_k^{-1}}_{(\theta/\alpha)R}(x_k - \alpha \mathsf{P}_k g_k)$. For $\mathsf{P} = \tau_t I$ this is exact: threshold $\theta\tau_t$, or a column-weighted SVT when one block mixes τ_t. For the circulant it has no closed form and needs inner iterations of FFTs and SVTs (no NUFFT).

  (b) Use a preconditioned primal–dual iteration (Condat–Vũ) with a fixed metric:

$$x_{k+1} = x_k - \mathsf{P}(\nabla f(x_k) + q_k), \qquad q_{k+1} = \operatorname{prox}_{\sigma(\lambda R)^*}\big(q_k + \sigma(2x_{k+1} - x_k)\big).$$

  It needs one gradient per iteration. Its fixed points satisfy $q = -\nabla f(x) \in \lambda \partial R(x)$, i.e. they minimise $f + \lambda R$ for every admissible (P, σ); admissibility is a step condition of the form $\mathsf{P}^{-1} - \sigma I \succ \frac{1}{2}\mathsf{N}$ [CHECK constants; designed, not implemented].

Temporal coupling: what a truncated temporal basis does to the conditioning

Fix an encode and a respiratory bin, and let $G = \operatorname{blockdiag}_f \mathsf{N}_f$ be the normal operator of the F cardiac frames. Write the frames as $x = (B \otimes I)c$ with $B \in \mathbb{C}^{F\times K}$, $B^{\mathsf{H}}B = I_K$ (e.g. the cyclic harmonics $|k| \le K_0$, $K = 2K_0 + 1$), and solve for c with the gradient $(B \otimes I)^{\mathsf{H}} G\big((B \otimes I)c\big) - (B \otimes I)^{\mathsf{H}} E^{\mathsf{H}} y$.

- *A unitary change of variables does nothing.* For $K = F$, $\hat{G} = (B \otimes I)^{\mathsf{H}} G (B \otimes I)$ is unitarily similar to G: same spectrum, same condition number, same iterates up to the transform.

- *Truncation does.* For $K < F$, $\hat{G}$ is a compression of G, so by Cauchy interlacing $\lambda_j(G) \le \lambda_j(\hat{G}) \le \lambda_{j+(F-K)n_v}(G)$: the spectrum can only move inwards. In the Fourier basis the blocks are

$$\hat{G}_{kk'} = \frac{1}{F}\sum_f \mathrm{e}^{-2\pi \mathrm{i}(k-k')f/F}\, \mathsf{N}_f, \qquad \hat{G}_{kk} = \bar{\mathsf{N}} := \frac{1}{F}\sum_f \mathsf{N}_f,$$

  i.e. every harmonic sees the pooled normal operator of all frames (about F times the samples of one frame), and the off-diagonal blocks are the Fourier coefficients of the frame-to-frame fluctuation of the sampling. Hence $\lambda_{\min}(\hat{G}) \ge \lambda_{\min}(\bar{\mathsf{N}}) - \|\hat{G} - I_K \otimes \bar{\mathsf{N}}\|$. A single frame's null space can only be reached by frame-localised functions, which need all F harmonics; with $K \ll F$ the unknowns per voxel drop from F to K against the same samples, an effective acceleration of $(K/F)R_{\text{frame}}$. How fast $\|\hat{G} - I_K \otimes \bar{\mathsf{N}}\|$ grows with K is not measured [CHECK].

- *The same estimator either way.* Projected gradient steps onto range$(B \otimes I)$ (P^2nP with $\mathcal{D} = BB^{\mathsf{H}}$, `--prior tlp`) and a Krylov or quasi-Newton solve in c have the same minimiser; the latter is the faster solver of it.

- *Cost.* A is untouched (one gradient per set, split over GPUs as now). B and B^{H} act pointwise in the voxels across all frames, so they run on voxel chunks of the whole stack: the projection BB^{H} over 280 sets of $4.17 \cdot 10^6$ voxels takes 2.0 s on the CPU [measured]. With the zeros of a block-diagonal basis skipped and one encode's voxel chunk of all frames in cache, Bc and $B^{\mathsf{H}}g$ are memory-bound at 50 GB/s: 0.27 s and 0.25 s per application at 2 mm for 40 frames and 18 coefficients, against $\approx$ 230 s for the gradient [measured, `hasty_recon::tsub`]. The state stays on the host, where the gradient streams the sets to the device(s) anyway, so no transposition between devices is needed. With the state sharded over devices by sets, every device would need all frames of its voxel range, i.e. a transpose of the (set, voxel) layout.

- *Secant pairs.* $B^{\mathsf{H}}\mathsf{N}B$ couples the coefficient sets that share a frame, so a per-set quasi-Newton metric (one secant pair per set, which assumes a block-diagonal Hessian) has to pair the coupled sets: one pair per connected component of the coefficient–frame incidence of B (per respiratory bin and encode for the cardiac bases). With one pair per coefficient set, ZMSHR1 diverged on a toy problem; with the groups it converges (`tests/tsub.rs`). The same holds for a temporal prior inside the gradient (Equation (36)).

- *Relation to Section 1.5.* The periodic B-splines $\beta^z_\kappa, \beta^v_\kappa$ are such a basis (in z linearly, in $\boldsymbol{v}$ through the phase), and the model evaluates $x_e(\cdot, t)$ at each sample's own time inside A, so encodes acquired at different times within a frame are not forced to share an image.

The prior inside the step: regularisation by denoising

In (31) only the data gradient moves the iterate; $\mathcal{D}$ moves it only towards its own range (for the temporal-mean projection of the static tissue that is a full step, the limit $\lambda \to \infty$ of a Newton step on $\frac{\lambda}{2}\|(I - \Pi)x\|^2$). Regularisation by denoising [8] puts the denoiser residual into the gradient,

$$x_{k+1} = x_k - \alpha \mathsf{P}_k\big(\nabla f(x_k) + \lambda(x_k - \mathcal{D}(x_k))\big), \tag{36}$$

so the secant pairs of ZMSHR1 contain the prior's curvature and the metric acts on both terms. When the residual is a gradient, the fixed points do not depend on P (unlike (35)):

- *Proximal denoisers.* For $\mathcal{D} = \mathrm{prox}_{\theta R}$ (e.g. unshifted LLR, R the sum of block nuclear norms), $x - \mathcal{D}(x) = \theta \nabla M_{\theta R}(x)$ with $M_{\theta R}$ the Moreau envelope, convex with a $1/\theta$-Lipschitz gradient. (36) is then preconditioned gradient descent on $f + \lambda\theta M_{\theta R}$, a smooth convex objective, and quasi-Newton theory applies as for f alone.

- *MMSE denoisers.* A denoiser trained with squared error at noise level σ satisfies Tweedie's formula $\mathcal{D}(x) = x + \sigma^2 \nabla \log p_\sigma(x)$ [11], so $x - \mathcal{D}(x) = -\sigma^2 \nabla \log p_\sigma(x)$ is the gradient of a potential (symmetric Jacobian), and (36) descends on $f - \lambda\sigma^2 \log p_\sigma$, a MAP objective with the smoothed prior [12].

- *Random block shifts.* Each shift is the prox of its own shifted block nuclear norm, so with a random shift per iteration the expected residual is the gradient of the shift-averaged envelope: a stochastic gradient of a fixed smooth convex objective.

- *Otherwise* (non-MMSE training, normalisation layers) no R with $\nabla R = x - \mathcal{D}(x)$ exists when the Jacobian is asymmetric [9], and (36) is a fixed-point equation without an objective. A gradient-step denoiser $\mathcal{D} = I - \nabla g_\vartheta$ [10] has an exactly conservative residual by construction.

Cost: one denoiser call per gradient, as in (31) [designed, not implemented; plan in `research/red_prior.md`].

Measured on ph0001 (2 mm, 10-min yarn scan)

- *Effective PSF of the pooled plain model* at an aortic point 8 voxels deep, $y = A\delta$, every second readout, the 12 strongest coils. With PCGLS and the circulant: $x(\boldsymbol{r}_0) = 0.024$, 0.506, 0.540, 0.561 at iterations 1, 10, 20, 100. The centre voxel then holds 56 % of the energy and $r \leq 1$ holds 84 %. The limit 0.56 is about the ball/cube ratio $\pi/6$ of a k-ball cut at Nyquist. Without a preconditioner: 0.000, 0.001, 0.005, 0.246 [measured].

- *Velocity phase of the time-averaged images* against the ideal frame-mean phase-contrast signal (deep lumen, voxels blood in every frame):

  - plain PCGLS with circulant on the outgoing half of the out-and-back readout: slope 0.96 (correlation 0.94);
  - adding the returning half (low $|\boldsymbol{\kappa}|$ again ≈ 9.6 ms after excitation): 0.75 (0.57);
  - the full model, 10 *unpreconditioned* pooled PCG iterations (`zmshr1s` run): 0.71 (0.90), and 0.41 at the wall [measured];
  - the same with the circulant metric (10 pooled iterations, relative residual 0.096): slope 0.88 (correlation 0.72) deep, but still 0.41 at the wall. The deep shrink is largely convergence; the wall shrink is not, and is specific to the full model (the plain model gives 0.68, its outgoing half 0.83) [measured; cause open].

Per-frame dynamics on ph0001 [measured]

Scored against the ideal per-frame phase-contrast signal (deep lumen, ≥ 5 voxels from the wall, blood in every frame), on the cardiac variation only: each voxel's temporal mean is removed from both sides, since a perfect time-averaged image already scores slope 0.43 and correlation 0.64 on the raw per-frame phases. The cardiac part has rms 0.40 rad.

- *Per-frame sampling.* One (frame, encode) set of the outgoing half holds $7.3 \cdot 10^5$ samples (the outgoing halves of 379 readouts), hitting 19% of the Nyquist cells of the k-ball; 31 virtual coils. Plain PCGLS with the circulant converges on it: relative residual $9 \cdot 10^{-3}$ after 25 iterations, $7 \cdot 10^{-4}$ after 100 (noise-free data).

- *Not noise.* The plain per-frame reconstruction (encodes of a rotation in one frame, outgoing half, 25 iterations) gives the same dynamics on noise-free and on noisy data: correlation 0.56 vs. 0.53, slope 0.68 vs. 0.67, 8 mm-ROI correlation 0.75 vs. 0.75. The per-frame error is deterministic.

- *Model mismatch, not undersampling.* On the noise-free data the dynamics are best after 25 iterations and then degrade while the residual keeps falling (slope / correlation / error 0.42 / 0.45 / 0.37 rad after 10, 0.68 / 0.56 / 0.37 after 25, 0.65 / 0.48 / 0.42 after 50, 0.50 / 0.34 / 0.50 after 100). For consistent data, PCGLS from $x_0 = 0$ decreases $\|x_k - x_{\text{true}}\|_{M^{-1}}$ monotonically (the CG error decrease, plus M^{-1}-orthogonality of range MA^{H} and $\ker A$). A phase score on the lumen voxels is not that norm, but a steady loss from 25 to 100 iterations on noise-free data indicates that the per-frame data are inconsistent with the plain model and least squares fits the inconsistency [CHECK: the same run on data synthesised with the reconstruction's own operator].

- *Readout motion* (the paragraph in Section 1.5). About TE, the reference of the encoding moments, the readout's first moment gives $\gamma|\boldsymbol{m}_1(\tau)| = 0.9$, 2.3 and 6.1 rad per m/s at 2.6,

3.6 and 5.5 ms ($|\boldsymbol{k}| = k_{\max}$ at the end of the outgoing half), and 7.0 on the return, against $\pi/1.5 = 2.1$ rad per m/s of velocity encoding. For blood at 1 m/s that is a 6 rad phase ramp across k-space, a velocity-dependent displacement of ≈ 4 mm, different in every cardiac frame. Its share is modest: the noise-free per-frame reconstruction on the first 1 / 2 / 4 ms of each readout ($\gamma|\boldsymbol{m}_1| \leq 0.9$ / 2.3 / 6.1 rad per m/s; the shorter windows also lose resolution) gives correlation 0.63 / 0.57 / 0.56 and error 0.32 / 0.36 / 0.37 rad (8 mm ROI: 0.77 / 0.76 / 0.75). Readout motion, B0 and fat during the readout are part of the inconsistency, not most of it.

- *LLR.* With $T = 280$ columns the $\kappa\hat{\sigma}$ rule keeps ≈ 200 components per block ($\hat{\sigma}$ lies inside the per-frame artefact spectrum). One LLR step on a per-frame start, with the rule varied: hard truncation to 8 components leaves most of the error (correlation 0.37 $\rightarrow$ 0.41); soft thresholding lowers it only by shrinking the dynamics as well (slope 0.46 $\rightarrow$ 0.37). The per-frame error lies in the leading temporal components of the 8^3 blocks, together with the dynamics.

- *Temporal.* The same error is incoherent across cardiac frames: projecting each voxel's series onto the cyclic harmonics $|k| \leq 4$ gives correlation 0.37 $\rightarrow$ 0.57 at unchanged slope (0.43). P^2nP with that projection as $\mathcal{D}$ reaches slope 0.45, correlation 0.57, error 0.32 rad after 5 iterations, then fits the per-frame error again (0.53, 0.45, 0.40 after 10).

- *Encodes sharing a frame.* Binning all encodes of a rotation into one frame shares their undersampling, but the velocity then refers to t_v with $(t_v - t_0)\, I \approx \mathrm{diag}(\Delta K^{+} D)$, $D_e = \boldsymbol{k}_e t_e - \boldsymbol{k}_0 t_0$: 1.35 TR after encode 0 for the referenced 7-point order, not the middle readout (a 0.44-frame lag, fixed). The off-diagonal of $\Delta K^{+} D$ (up to 3.1 TR) mixes the other axes' acceleration into each component: 0.09 m/s rms against 0.21 m/s of cardiac dynamics. Over all 7! orders, the reference in the middle with symmetric axes reduces it eightfold (0.011 m/s).

Lagged correction. $\tilde{g}_k = \mathsf{N}_{\mathrm{s}}(x_k - x_a) + g(x_a)$, refreshing $x_a \leftarrow x_k$ when $\|x_k - x_a\| > \tau \|x_k\|$. This removes the surrogate bias at the fixed point. Cheap-only gradients are unsafe under a quasi-Newton metric: secant pairs from a mismatched operator act where N has small eigenvalues. 16^3 toy, $\Phi_{\max} = 3$, 100 iterations, identity denoiser; cost in plain-Toeplitz units, exact = 39 [measured]:

gradient	NRMSE	exact applications	cost
exact every iteration	0.213	100	15600
plain surrogate only	1.29	0	400
Weyl order 2 only	$7.7 \cdot 10^4$	0	12800
lagged plain, $\tau = 0.1$ / 0.05	0.218 / 0.214	12 / 19	2320 / 3440

This is 4.5–6.7$\times$ cheaper at equal error on the toy. With the reduced-segment surrogate on the real operator the gain is smaller and needs a mismatch $\lesssim 3\,\%$ (Section 1.6). It is not yet tested with the real denoiser.

1.9 Validation and limits

Truth data come from a GPU direct sum over a fine point cloud with exact (M_0, M_1, M_2), susceptibility B0, GNL and the full concomitant phase. It shares nothing with the operators and agrees with an f64 CPU sum to $1.2 \cdot 10^{-5}$ [measured]:

experiment	result
time-averaged LS per encode (R4, 60 s): no physics / B0 / B0+conc.	0.211 / 0.249 / 0.131 m/s
CFD case (RACLETTE n001), B0 + conc. + GNL, model-exact data	image $4.3 \cdot 10^{-3}$, velocity 2.4 mm/s
velocity model (Section 1.5) vs. complex + LLR	0.138 vs. 0.183 m/s

Limits.

- Concomitant fields are first order in G/B_0, computed from nominal or GIRF-filtered waveforms; eddy-current residuals have not been measured.
- Taylor stencils hold for $|\chi_j| \lesssim 0.7$ rad. At 0.55 T they need least-squares stencils inside the segments, which are not built.
- L grows with $\Delta f\, T_{\mathrm{ro}}$; long readouts pay directly in FFTs.
- Readout motion and intravoxel dephasing are not modelled.
- Operator error on typical images is not the relevant accuracy for a quasi-Newton iteration (Section 1.8).

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